

Piyush Hinduja

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Portfolio: <https://piyushinduja.github.io/Portfolio/> | LinkedIn: [linkedin.com/in/piyush-hinduja/](https://www.linkedin.com/in/piyush-hinduja/)

EDUCATION

The University of Utah, Salt Lake City, Utah

August 2023 - May 2025

Master of Science (MS) Computer Science

GPA: 3.9/4.0

University of Mumbai, Bandra, India

July 2019 - May 2023

Bachelor of Engineering (BE) Computer Engineering

GPA: 3.84/4.0

SKILLS

Languages: Proficient (4-5 yrs): Python, Java, JavaScript Intermediate (3 yrs): C, C++ Beginner (1 yr): Go, Kotlin, Dart, SQL, Shell

Machine Learning and Data Processing: PyTorch, HuggingFace, TensorFlow, Langchain, Scikit-Learn, Pandas, PySpark, OpenCV

DevOps & Databases: AWS, Amazon RDS, CloudLab, Docker, Kubernetes, MySQL, MongoDB, ChromaDB, Git, JIRA

Web Development: Node.js, React.js, TypeScript, HTML, XML, CSS

Core Computer Science: Data Structures (Trees, Graphs, Linked Lists, Hashmaps), Algorithms (Recursion, DP, Searching, Sorting), CI/CD

WORK EXPERIENCE

Data Analyst | The University of Utah, Salt Lake City, Utah

March 2024 - Present

- Architecting a model to **track changes in 10-K filings** over time, analyzing forward-looking intensity, document length, and financial tone based on insights from **Muslu et al.'s 2015 study**.
- Improved data preprocessing, boosting **forward-looking statement identification by 27%** and **reducing per-filing processing time by 40%**.
- Implementing **AI-driven analysis** by prompting ChatGPT, comparing AI-generated outputs against manually coded algorithms to measure performance and efficiency.
- Expanding work to include similar analysis on **10-Q and 8-K filings, and conference calls**, aiming to **broaden the scope of financial analysis** and improve real-time insights.

Machine Learning Researcher | FLUX Research Group, Salt Lake City, Utah

August 2024 - Present

- Architected and implemented a comprehensive CloudLab profile enabling seamless **integration of NVIDIA drivers and CUDA toolkit**, **reducing environment setup time by 90%** and benefiting 50+ researchers across multiple projects.
- Collaborating with the Wireless Powder Testbed team to implement **predictive modeling solutions** focusing on enhancing experimental outcomes through machine learning applications for a **network of 10000+ UEs** (User Equipments).

Full Stack Developer Intern | Exposys Data Labs, Bangalore, India

June 2021 - July 2021

- Spearheaded the launch of a **tourism project**, integrating key features to **enhance user experience** and deliver valuable information about 50+ Indian travel destinations.
- Utilized **MERN stack** for an efficient frontend and backend integration, creating a **responsive, user-friendly interface** and enabling content management.

PROJECTS

Road Damage Detection | (~120 hours) | Object-Detection, YOLOv5, FasterRCNN

[GitHub Link](#)

- Collaborated with a team of four to develop a road damage detection system using **YOLOv5** and **FasterRCNN**, integrating a user-friendly web interface (deployed via **Streamlit**), leveraging the **RDD-2020 dataset**.
- Achieved model accuracies of **52.67%** with YOLOv5 and **44.45%** with FasterRCNN by training and **fine-tuning** on different **hyper-parameters**.
- The system plays a vital role, delivering a **70% improvement in road damage detection accuracy** over manual methods, helping **lower the frequency of accidents and fatalities**.

Dependency Parser | (~60 hours) | PyTorch, GloVe Embeddings

[GitHub Link](#)

- Streamlined a Transition based Dependency parser for the **university's NLP group**, inspired by "A Fast and Accurate Dependency Parser using Neural Networks" (**Chen and Manning, 2014**), returning a parse tree for a sentence in less than **100 ms**.
- Focused on Data Transformation, Model Building and Fine-tuning over different embeddings and hyperparameters, achieving a **UAS (Unlabeled Attachment Score) of 0.76** and **LAS (Labeled Attachment Score) of 0.705**.

Pizzeria | (~60 hours) | Web Development, MERN, Firebase

[GitHub Link](#)

- Designed a **pizza ordering web application** featuring live order updates, personalized carts, favorite pizza options, and an integrated chat feature for **real-time customer-store communication**.
- Utilized a tech stack including **HTML5, CSS, JavaScript, Node.js, Express.js, MongoDB**, and libraries such as **Socket.io** for real-time updates and **Passport** for secure authentication.

Language Modeling | (~40 hours) | NLP, RNN, N-gram, PyTorch

[GitHub Link](#)

- Developed an **LSTM-based RNN** to predict character sequences, enhancing **probabilistic language modeling**, and achieving a **perplexity measure of 9.43**.
- Employed **teacher forcing** during training and evaluated the model using the perplexity metric.
- Engineered a **N-gram statistical model** incorporating **Laplace smoothing** to mitigate zero probability issues, and achieving a **perplexity of 17.85**.